

Youngmin Kim

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Professional Summary

*A Ph.D. Student in Artificial Intelligence with a research focus on **video understanding** and **human-interacted AI**. My work investigates how AI systems can interpret complex visual information from videos and connect this understanding to human perception and communication, with particular interest in **nonverbal expressions** and **sign language**.*

Education

Yonsei University	Seoul, South Korea
Ph.D. Student in Artificial Intelligence	Sep. 2023 – Present
Incheon National University	Incheon, South Korea
B.S. in Economics & Computer Science	Mar. 2016 – Mar. 2023

Publications (*: Equal Contribution)

Conference Publications

1. **QWERTY: Training-Free Motion Control via Query-Warped Video Diffusion Transformers** [paper]
Kyobin Choo, **Youngmin Kim**, Hyunkyung Han, Geunrip Park, Chanyoung Kim, Sunyoung Jung, Seong Jae Hwang
European Conference on Computer Vision (ECCV), 2026
2. **ViKey: Enhancing Temporal Understanding in Videos via Visual Prompting** [paper]
Yeonkyung Lee*, Dayun Ju*, **Youngmin Kim**, Seil Kang, Seong Jae Hwang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
3. **MAVL: A Multilingual Audio-Video Lyrics Dataset for Animated Song Translation** [paper]
Woohyun Cho, **Youngmin Kim**, Sunghyun Lee, Youngjae Yu
Empirical Methods in Natural Language Processing (EMNLP), 2025
4. **Speaking Beyond Language: A Large-Scale Multimodal Dataset for Learning Nonverbal Cues from Video-Grounded Dialogues** [paper]
Youngmin Kim*, Jiwan Chung*, Jisoo Kim, Sunghyun Lee, Sangkyu Lee, Junheok Kim, Cheoljong Yang, Youngjae Yu
Association for Computational Linguistics (ACL), 2025
5. **Scalp Diagnostic System With Label-Free Segmentation and Training-Free Image Translation** [paper]
Youngmin Kim*, Saejin Kim*, Hyeon Moon, Youngjae Yu, Junhyug Noh
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2025
6. **A 2-Stage Model for Vehicle Class and Orientation Detection with Photo-Realistic Image Generation** [paper]
Youngmin Kim*, Donghwa Kang*, Hyeongboo Baek
IEEE International Conference on Big Data (IEEE BigData), 2022
7. **Pedestrian Accident Prevention Model Using Deep Learning and Optical Flow** [paper]
Youngmin Kim*, Donghwa Kang*, Hyeongboo Baek
Korea Computer Congress (KCC), 2021
Best Paper Award (Undergraduate Session)

Journal Publications

1. **AI-Powered Real-Time System for Automated Concrete Slump Prediction via Video Analysis** [paper]
Youngmin Kim, Giyeong Oh, Kwangsoo Youm*, Youngjae Yu*
Automation in Construction (AiC), 2026, IF: 11.5
2. **Preprocessing for Keypoint based Sign Language Translation without Glosses** [paper]

Youngmin Kim, Hyeongboo Baek.
Sensors, 2023, IF: 3.8

3. **A Study of Tram-Pedestrian Collision Prediction Method Using YOLOv5 and Motion Vector** [paper]
Youngmin Kim, Hyeonuk Ahn, Heegyun Jeon, Jinpyung Kim, Gyujin Jang, Hyeonchyeol Hwang
Korea Information Processing Society (KIPS), 2021
4. **Optical Flow Estimation Techniques and Recent Research Trends** [paper]
Youngmin Kim, Hyeonuk Ahn, Jinpyung Kim
Korea Information Processing Society Reviews, 2021

Pre-prints

1. **Towards Continuous Sign Language Conversation from Isolated Signs** [paper]
Youngmin Kim, Kyobin Choo, Jiwoo Park, Minseo Kim, Chanyoung Kim, Junhyeok Kim, Seong Jae Hwang
Pre-print
2. **Sparse Feature Analysis and Control for Robust Vision-Language-Action Models** [paper]
Minseo Kim, Junhyeok Kim, Jinyeong Kim, **Youngmin Kim**, Seong Jae Hwang
Pre-print
3. **Attention Weight as a Hierarchical Tree in ViT** [paper]
Junhyeok Kim, Minseo Kim, Seil Kang, **Youngmin Kim**, Seong Jae Hwang
Pre-print

Experience

MIR Lab, Yonsei University

Undergraduate Researcher

Mar. 2022 – Mar. 2023

RAISE Lab, Incheon National University

Undergraduate Researcher

Dec. 2021 – Feb. 2023

CVAI Lab, Advanced Institute of Convergence Technology (AICT)

Research Assistant

Sep. 2020 – Aug. 2021

Scholarly Activities

Invited Speaker

- **Samsung AI Education** 2023
Introduction to Multimodal Research

Teaching Assistant

- Introduction to AI Research (Yonsei University) 2024
- Reinforcement Learning (Yonsei University) 2024
- Basic of PyTorch (Upstage) 2023

Academic Service

Journal Reviewer

- IEEE Transactions on Image Processing

Selected Projects

LG Electronic

Multimodal Contents Recommendation System

Mar. 2025 – Aug. 2025

GS E&C

Video-based Slump Prediction

Dec. 2023 – Feb. 2025

NCSOFT

Multimodal Conversational Agent

Jan. 2024 – Jun. 2024

Awards

AI-Hub Idea Challenge Competition 2021

1st prize, hosted by South Korea's National Information Society Agency (NIA)

KCC 2021 Undergraduate Paper Competition 2021

1st prize, hosted by Korean Institute of Information Scientists and Engineers (KIISE)

KED 2021 Industrial Innovation Big Data Platform Competition 2021

2nd prize, hosted by Korea Enterprise Data (KED)

Patents

Method and apparatus for avoiding collision between vehicle and object, computer program 2023

KR102533419B1

Electronic apparatus and method for searching distress, unmanned aerial vehicle, computer program 2023

KR102587311B1

Apparatus and method for analyzing data, apparatus and method for predicting abnormality, computer program
2023

KR102480394B1